Quantification: 876_z89c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2175.11	2169	869.225	px^2
major_axis_length_px	lacuna shape	84.7631	81.6741	19.9666	px
minor_axis_length_px	lacuna shape	33.7022	35.8518	7.7268	px
aspect_ratio	lacuna shape	2.557	2.7987	0.4604	unitless
eccentricity	lacuna shape	0.9107	0.934	0.04	unitless
solidity	lacuna shape	0.8171	0.8169	0.0544	unitless
roots_count	headline	12.6667	12	5.1962	count
ring_length_r30_px	headline	756.778	728	63.5842	px
ring_length_r60_px	reported	1955.67	2026	306.631	px
owned_length_px	ownership-dependent	7127.47	7437.8	2984.2	px
edge_count	ownership-dependent	532.444	569	217.196	count
mean_edge_length_px	ownership-dependent	13.327	13.2642	0.3079	px
perimeter_px	normalised	294.887	274.676	88.6524	px
ring_area_r30_px2	normalised	9333.67	9180	1991.24	px^2
ring_area_r60_px2	normalised	23113.9	23485	4641.06	px^2
in_frame_fraction_r30	normalised	0.9828	1	0.0515	unitless
in_frame_fraction_r60	normalised	0.9539	1	0.0857	unitless
ring_density_r30	normalised	0.0836037	0.0825109	0.01426	px^-1
ring_density_r60	normalised	0.0857258	0.0837235	0.00993884	px^-1
roots_per_100px_perimeter	normalised	4.2708	4.2819	1.1366	per 100 px
ring_attached_length_r30_px	network v2	711.667	695	76.6714	px
ring_attached_length_r60_px	network v2	1821.44	1742	356.003	px
ring_length_w_r30_px	network v2	854.787	829.914	69.0295	px
ring_length_w_r60_px	network v2	2218.82	2318.52	347.997	px
sholl_crossings_r10	network v2	17.1111	17	2.5712	count
sholl_crossings_r20	network v2	20.6667	20	2.5981	count
sholl_crossings_r30	network v2	22.8889	22	4.4001	count
loop_count	loops, report only	89.1111	99	38.1361	count
loop_edge_share	loops, report only	0.584	0.5757	0.079	share
ring_width_mean_r30_px	width	5.9701	5.6859	0.6517	px

Per field
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measure	value	units
analysed_area_px2	1.02251e+06	px^2
total_skeleton_length_px	77693	px
canalicular_length_density_per_px	0.0759826	px^-1
junction_count	2880	count
junction_density_per_px2	0.0028166	px^-2
lacuna_count	10	count

lacunae_per_px2	9.78e-06	px^-2
skeleton_component_count	532	count
mean_component_length_px	146.04	px
median_component_length_px	10	px
field_length_density_w_per_px	0.0856357	px^-1
field_density_without_flagged_per_px	0.076949	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	90	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0759826	px^-1
field_length_density_with_unattached_per_px	0.0760707	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	982	count
loop_edge_share	0.5431	share
edge_count	6093	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2888	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	63249	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6490	249.288	41.8559	5.9559	0.9858	0.5841	-1.5551	22.8282	886.791	True
2	3313	115.919	39.5634	2.93	0.94	0.7245	-1.5598	154.062	372.366	False
3	2754	105.554	35.8518	2.9442	0.9406	0.7558	1.5255	165.057	102.925	False
4	3301	102.585	44.3824	2.3114	0.9016	0.8169	-1.4051	307.062	909.902	False
5	2169	78.4025	38.343	2.0448	0.8723	0.8413	1.557	311.008	321.631	False
6	956	61.2609	21.1134	2.9015	0.9387	0.8	1.5273	461.978	61.1475	False
7	2610	87.8807	38.6915	2.2713	0.8979	0.8782	-1.531	535.001	198.764	False
8	1500	59.3512	33.3189	1.7813	0.8276	0.8966	1.5421	545.937	440.713	False
9	1670	81.6741	26.9572	3.0298	0.944	0.8337	-1.4297	692.65	612.224	False
10	1303	70.2403	25.0978	2.7987	0.934	0.8068	-1.3129	914.738	39.4635	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	34	1057	2018	5611.78	391	14.3524	
2	23	870	2363	9691.35	700	13.8448	5.2883
3	15	780	1854	4151.46	314	13.2212	5.6859
4	16	711	2116	11987.5	892	13.4389	5.6757
5	12	845	2136	5509.7	411	13.4056	6.556
6	12	728	1645	3767.59	293	12.8587	6.9412
7	14	761	2196	7437.8	569	13.0717	6.5917

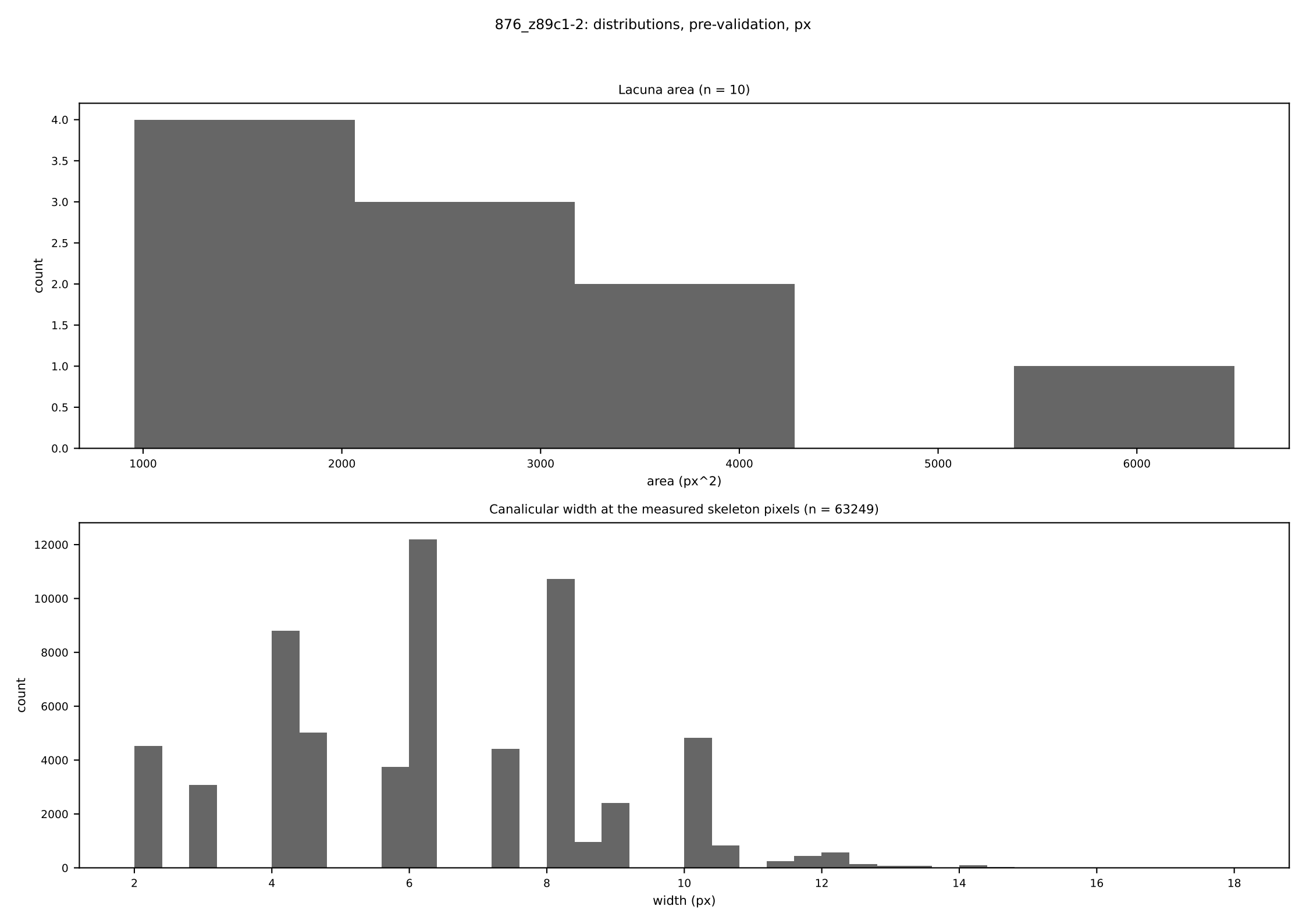
8	6	703	2026	8738.52	665	13.1406	6.372
9	9	719	1906	9136.12	667	13.6973	5.4829
10	7	694	1359	3727.24	281	13.2642	5.1371

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	1053.68	14114	25792	0.6408	0.5406	0.0748902	0.0782413	3.2268
2	423.014	12381	29774	1	1	0.070269	0.0793646	5.4372
3	413.328	11320	26755	1	0.9628	0.0689046	0.0692955	3.6291
4	373.665	11365	27711	1	0.9875	0.0625605	0.0763596	4.2819
5	273.505	9180	23731	1	1	0.0920479	0.0900089	4.3875
6	191.315	7221	16849	1	0.887	0.100817	0.0976319	6.2724
7	277.848	9459	23485	1	1	0.0804525	0.0935065	5.0387
8	190.593	7507	20585	1	1	0.0936459	0.0984212	3.1481
9	274.676	8714	22903	1	1	0.0825109	0.0832205	3.2766
10	236.043	6856	16232	0.8455	0.748	0.101225	0.0837235	2.9656

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	1037	1910	1178.44	2232.76	29	19	22
2	797	2280	966.482	2647.2	19	22	28
3	720	1660	876.24	2105.76	21	20	22
4	565	1742	794.784	2396.9	14	22	27
5	834	2083	958.411	2421.43	20	21	22
6	689	1554	829.914	1850.24	18	18	22
7	739	2190	873.068	2516.82	17	20	26
8	695	2003	800.985	2318.52	16	17	22
9	680	1734	807.257	2174.31	14	26	24
10	686	1147	785.943	1538.24	15	20	13


876_z89c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 63249)
count
width (px)